GOA COLLEGE OF ENGINEERING

"BHAUSAHEB BANDODKAR TECHNICAL EDUCATION COMPLEX"
FARMAGUDI, PONDA- GOA - INDIA

MATHEMATICS-III
UNIT 1: MATRICES
LECTURE 4

DEVELOPED BY MATHEMATICS FACULTY
DEPARTMENT OF SCIENCE & HUMANITIES
GOA COLLEGE OF ENGINEERING

MATRICES LECTURE 4

TOPICS TO BE COVERED:

- Elementary Transformations.
- Inverse of the Matrix using Elementary Transformations.
- Echelon Form of the Matrix.
- Normal Form of the Matrix.
- Rank of the Matrix.

Elementary Transformations

- Any one of the following operations on a matrix is called an Elementary Transformation.

1. Interchanging any two rows(or columns).This transformations is denoted by R_i<—>Rj(C_i<—>Cj)(R_{ij})(C_{ij})if the i^{th} *and* j^{th} *rows/columns are interchanged*.
2. Multiplication of the elements of any row(or column) by a non-zero scalar quantity to get R_i—> kR_i (C_i—> kC_i)(kR_i)(R_i(k))which is multiplying k to the i^{th} row/column.
3. Addition of constant multiplication of the elements of any row/column to the corresponding element of any other row/column. R_i—> R_i+ kR_j (C_i—> C_i+ kC_j).
4. Note: Two matrices are said to be equivalent if one is obtained from other by elementary Transformations. It is denoted by A~B, where B is obtained from A by using elementary transformation.

Find the resultant matrix after using following Elementary Tranformations on the matrix $A = \begin{bmatrix} 1 & -2 & 3 \\ 0 & 1 & 2 \\ 3 & 4 & 1 \end{bmatrix}$ i) $R_1 \rightarrow R_2$ ii) $R_1 \rightarrow 5R_1$ iii) $R_2 \rightarrow R_2 + 3R_1$

Given $A = \begin{bmatrix} 1 & -2 & 3 \\ 0 & 1 & 2 \\ 3 & 4 & 1 \end{bmatrix}$

i) $\begin{bmatrix} 1 & -2 & 3 \\ 0 & 1 & 2 \\ 3 & 4 & 1 \end{bmatrix} \sim \begin{bmatrix} 0 & 1 & 2 \\ 1 & -2 & 3 \\ 3 & 4 & 1 \end{bmatrix}$ $R_1 \leftrightarrow R_2$

ii) $\begin{bmatrix} 1 & -2 & 3 \\ 0 & 1 & 2 \\ 3 & 4 & 1 \end{bmatrix} \sim \begin{bmatrix} 5 & -10 & 15 \\ 0 & 1 & 2 \\ 3 & 4 & 1 \end{bmatrix}$ $R_1 \rightarrow 5R_1$

iii) $\begin{bmatrix} 1 & -2 & 3 \\ 0 & 1 & 2 \\ 3 & 4 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & -2 & 3 \\ 3 & -5 & 11 \\ 3 & 4 & 1 \end{bmatrix}$ $R_2 \rightarrow R_2 + 3R_1$

Elementary Matrix

- A matrix obtained from a unit matrix by single elementary transformation is called an Elementary Matrix.

Ex. If $I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ consider the matrix obtained by $R_3 —> R_3 + 3R_2$

$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 3 & 1 \end{bmatrix}$ is called the Elementary Matrix.

Echelon Form of a Matrix

The matrix is said to be Echelon Matrix if

- all rows consisting of only zeroes are at the bottom.
- the leading coefficient (also called the pivot) of a nonzero row is always strictly to the right of the leading coefficient of the row above it.

Ex. $A = \begin{bmatrix} 1 & 2 & 0 & 5 \\ 0 & 3 & 2 & 1 \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ $B = \begin{bmatrix} 1 & 2 & 3 & 5 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 5 \end{bmatrix}$

Note : The Echelon form of square matrix is looks like a Upper Triangular Matrix.

Echelon form of a matrix

- Reduce matrix $\begin{bmatrix} 1 & 2 & -2 & -3 \\ -3 & 2 & 3 & 1 \\ -2 & 4 & 1 & -2 \end{bmatrix}$ to its equivalent Echelon matrix.

- $\begin{bmatrix} 1 & 2 & -2 & -3 \\ -3 & 2 & 3 & 1 \\ -2 & 4 & 1 & -2 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & -2 & -3 \\ 0 & 8 & -3 & -8 \\ 0 & 8 & -3 & -8 \end{bmatrix}$ $R_2 \text{—>} R_2\text{+}3R_1$, $R_3 \text{—>} R_3\text{+}2R_1$

- $\begin{bmatrix} 1 & 2 & -2 & -3 \\ 0 & 8 & -3 & -8 \\ 0 & 8 & -3 & -8 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & -2 & -3 \\ 0 & 8 & -3 & -8 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ $R_3 \text{—>} R_3\text{-}R_2$

which is the echelon form equivalent to the above matrix.

Inverse of matrix(Elementary Transformation) (Gauss-Jordan Method)

If A is reduced to I by elementary transformation then

$$PA = I \quad where\ P = P_n P_{n-1} \dots . P_2 P_1 \text{ =Elementary Matrix.}$$

$$\boldsymbol{P = A^{-1}}$$

Working Rule:

Write $A = IA$.

Perform elementary row transformation on A of the left side and on I of the right hand side so that A is reduced to I and I of right hand side is reduced to P getting $\mathrm{I} = \mathrm{PA}$.

Then P is the inverse of A.

Example(Gauss Jordan Method)

$$Find\ the\ Inverse\ of\ the\ Matrix\ A = \begin{bmatrix} 1 & -1 & 1 \\ 4 & 1 & 0 \\ 8 & 1 & 1 \end{bmatrix}$$

Solution:

$$A = IA.$$

$$\begin{bmatrix} 1 & -1 & 1 \\ 4 & 1 & 0 \\ 8 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A$$

$$R_2 \text{—>} R_2 - 4R_1, R_3 \text{—>} R_3 - 8R_1$$

$$\begin{bmatrix} 1 & -1 & 1 \\ 0 & 5 & -4 \\ 0 & 9 & -7 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -4 & 1 & 0 \\ -8 & 0 & 1 \end{bmatrix} A$$

$$R_2 \rightarrow \frac{1}{5} R_2$$

$$\begin{bmatrix} 1 & -1 & 1 \\ 0 & 1 & -4/5 \\ 0 & 9 & -7 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -4/5 & 1/5 & 0 \\ -8 & 0 & 1 \end{bmatrix} A$$

$$R_1 \rightarrow R_1 + R_2, R_3 \rightarrow R_3 - 9R_2$$

$$\begin{bmatrix} 1 & 0 & 1/5 \\ 0 & 1 & -4/5 \\ 0 & 0 & 1/5 \end{bmatrix} = \begin{bmatrix} 1/5 & 1/5 & 0 \\ -4/5 & 1/5 & 0 \\ -4/5 & -9/5 & 1 \end{bmatrix} A$$

$$R_3 \rightarrow 5R_3$$

$$\begin{bmatrix} 1 & 0 & 1/5 \\ 0 & 1 & -4/5 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1/5 & 1/5 & 0 \\ -4/5 & 1/5 & 0 \\ -4 & -9 & 5 \end{bmatrix} A$$

$$R_1 \to R_1 - \frac{1}{5}R_3, R_2 \to R_2 + \frac{4}{5}R_3$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 & -1 \\ -4 & -7 & 4 \\ -4 & -9 & 5 \end{bmatrix} A$$

Therefore, $\boldsymbol{A^{-1}} = \begin{bmatrix} \mathbf{1} & \mathbf{2} & \mathbf{-1} \\ \mathbf{-4} & \mathbf{-7} & \mathbf{4} \\ \mathbf{-4} & \mathbf{-9} & \mathbf{5} \end{bmatrix}$

A. $\boldsymbol{A^{-1}} = \begin{bmatrix} 1 & -1 & 1 \\ 4 & 1 & 0 \\ 8 & 1 & 1 \end{bmatrix} \begin{bmatrix} \mathbf{1} & \mathbf{2} & \mathbf{-1} \\ \mathbf{-4} & \mathbf{-7} & \mathbf{4} \\ \mathbf{-4} & \mathbf{-9} & \mathbf{5} \end{bmatrix} = \begin{bmatrix} \mathbf{1} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{1} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{1} \end{bmatrix}$

Normal Form(Canonical Form)

A matrix is said to be in normal form if it is in any one of the in any one of the states given below:

i)I_r i:e identity matrix of order r.

$ii)[I_r \quad \mathbf{0}]$ of the type $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$

$iii)\begin{bmatrix} I_r \\ \mathbf{0} \end{bmatrix}$ of the type $\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$

$iv)\begin{bmatrix} I_r & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{bmatrix}$ of the type $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

Example(Normal Form)

Every matrix is equivalent to a matrix in normal form. i:e every matrix can be reduced to a matrix of normal form by using elementary operations.

$$Reduce\ to\ normal\ form\ the\ following\ matrix\ A=\begin{bmatrix}2 & 1 & -3 & 6\\ 3 & -3 & 1 & 2\\ 1 & 1 & 1 & 2\end{bmatrix}$$

Solution:

$$A=\begin{bmatrix}2 & 1 & -3 & 6\\ 3 & -3 & 1 & 2\\ 1 & 1 & 1 & 2\end{bmatrix}\sim\begin{bmatrix}1 & 1 & 1 & 2\\ 3 & -3 & 1 & 2\\ 2 & 1 & -3 & 6\end{bmatrix} R_1 <\!\!-\!\!> R_3$$

$$A\sim\begin{bmatrix}1 & 1 & 1 & 2\\ 0 & -6 & -2 & -4\\ 0 & -1 & -5 & 2\end{bmatrix} R_2 -\!\!> R_2-3R_1, R_3 -\!\!> R_3-2R_1$$

$$A\sim\begin{bmatrix}1 & 0 & 0 & 0\\ 0 & -6 & -2 & -4\\ 0 & -1 & -5 & 2\end{bmatrix} C_2 -\!\!> C_2-C_1, C_3 -\!\!> C_3-C_1, C_4 -\!\!> C_4-2C_1$$

$$R_2 <\!\!-\!\!> R_3 \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -6 & -2 & -4 \\ 0 & -1 & -5 & 2 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & -5 & 2 \\ 0 & -6 & -2 & -4 \end{bmatrix}$$

$$R_2 \!-\!\!> (-1)R_2 \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & -5 & 2 \\ 0 & -6 & -2 & -4 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 5 & -2 \\ 0 & -6 & -2 & -4 \end{bmatrix}$$

$$R_3 \!-\!\!> R_3 + 6R_2 \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 5 & -2 \\ 0 & -6 & -2 & -4 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 5 & -2 \\ 0 & 0 & 28 & -16 \end{bmatrix}$$

$$C_3 \!-\!\!> C_3 - 5C_2, C_4 \!-\!\!> C_4 + 2C_2 \qquad A \sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 28 & -16 \end{bmatrix}$$

$$A \sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -16/28 \end{bmatrix} R_3 \to \frac{1}{28} R_3$$

$$A \sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} C_4 \to C_4 + \frac{16}{28} C_3$$

Therefore, $A \sim [I_3 \quad 0]$ this is the normal form of A.

- $\boldsymbol{Find\ non\ singular\ matrices\ P\ and\ Q\ such\ that\ PAQ\ is\ in\ normal\ form\ where, A = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{bmatrix}}$.

- Solution:

Order of A is 2 X 3.

As number of rows in A is 2 and number of columns in A is 3,therefore we consider $I_2\ and\ I_3$.

$$Consider\ \ A = I_2 A I_3$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} A \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Now use elementary transformations on A to reduce the matrix A into its Normal Form.

Every row transformation apply on I_2 and every column transformation apply on $I_3\ so\ that\ I_2$ and I_3 converted into two non-singular matrices P and Q respectively such that PAQ is in normal form.

$$\begin{bmatrix}1 & 2 & 3\\3 & 1 & 2\end{bmatrix} = \begin{bmatrix}1 & 0\\0 & 1\end{bmatrix} A \begin{bmatrix}1 & 0 & 0\\0 & 1 & 0\\0 & 0 & 1\end{bmatrix} \; R_2 \text{—>} R_2 - 3R_1$$

$$\begin{bmatrix}1 & 2 & 3\\0 & -5 & -7\end{bmatrix} = \begin{bmatrix}1 & 0\\-3 & 1\end{bmatrix} A \begin{bmatrix}1 & 0 & 0\\0 & 1 & 0\\0 & 0 & 1\end{bmatrix} \; R_2 \text{—>} R_2 - 3R_1$$

$$\begin{bmatrix}1 & 0 & 0\\0 & -5 & -7\end{bmatrix} = \begin{bmatrix}1 & 0\\-3 & 1\end{bmatrix} A \begin{bmatrix}1 & -2 & -3\\0 & 1 & 0\\0 & 0 & 1\end{bmatrix} \; C_2 \text{—>} C_2 - 2C_1, C_3 \text{—>} C_3 - 3C_1$$

$$\begin{bmatrix}1 & 0 & 0\\0 & 1 & 7/5\end{bmatrix} = \begin{bmatrix}1 & 0\\3/5 & -1/5\end{bmatrix} A \begin{bmatrix}1 & -2 & -3\\0 & 1 & 0\\0 & 0 & 1\end{bmatrix} \; R_2 \text{—>-} \frac{1}{5} R_2$$

$$\begin{bmatrix}1 & 0 & 0\\0 & 1 & 0\end{bmatrix} = \begin{bmatrix}1 & 0\\3/5 & -1/5\end{bmatrix} A \begin{bmatrix}1 & -2 & -1/5\\0 & 1 & -7/5\\0 & 0 & 1\end{bmatrix} \; C_3 \text{—>} C_3 - \frac{7}{5} C_2$$

So there exist P= $\begin{bmatrix} 1 & 0 \\ 3/5 & 1/5 \end{bmatrix}$ and Q= $\begin{bmatrix} 1 & -2 & -1/5 \\ 0 & 1 & -7/5 \\ 0 & 0 & 1 \end{bmatrix}$ both are non singular such that PAQ is in normal form.

$$\boldsymbol{PAQ} = [\boldsymbol{I_2} \quad \mathbf{0}]$$

Sub Matrix of a Matrix

Definition: A sub matrix of a matrix is the matrix obtained by choosing a subset of rows and columns of the matrix and forming a new matrix by choosing those entries in the same position that appear in both the rows and columns of those selected. For example for the

Matrix A= $\begin{bmatrix} \mathbf{-1} & \mathbf{2} & \mathbf{3} & \mathbf{-2} \\ \mathbf{2} & \mathbf{-5} & \mathbf{1} & \mathbf{2} \\ \mathbf{3} & \mathbf{-8} & \mathbf{5} & \mathbf{2} \\ \mathbf{5} & \mathbf{-12} & \mathbf{-1} & \mathbf{6} \end{bmatrix}$ the matrix $\begin{bmatrix} 2 & -5 & 1 \\ 3 & -8 & 5 \end{bmatrix}$ is a submatrix of

obtained by taking the elements common to the columns C_1, C_2, C_3 and rows R_2, R_3.

Rank of the Matrix

Definition : The rank of the matrix is said to be " r" if

i)It has at least one non-singular sub matrix of order r.

ii)Every square sub matrix of order greater than r is singular.

Rank of the matrix = Number of non zero rows in Echelon form.

If the matrix A is equivalent to any one of the normal form

$i)\, I_r \quad ii)\, [I_r \quad 0] \quad iii)\, \begin{bmatrix} I_r \\ 0 \end{bmatrix} \quad iv)\, \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}$

Then **Rank(A)= r**

Examples

a) $A = \begin{bmatrix} 1 & 2 & 3 \\ -4 & 0 & 5 \end{bmatrix}$, $\begin{vmatrix} 1 & 2 \\ -4 & 0 \end{vmatrix}$=-8≠0 Therefore rank(A)=2.

b) $A = \begin{bmatrix} 1 & 1 & 2 \\ 2 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$ |A|=3≠0 Rank A=3.

c) $A = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 2 & 1 & 1 & 1 \\ 1 & 1 & 0 & 1 \end{bmatrix}_{3\times 4}$

$\begin{vmatrix} 1 & 0 & 1 \\ 2 & 1 & 1 \\ 1 & 1 & 0 \end{vmatrix}$=0, $\begin{vmatrix} 1 & 1 & 0 \\ 2 & 1 & 1 \\ 1 & 0 & 1 \end{vmatrix}$=0, $\begin{vmatrix} 1 & 0 & 0 \\ 2 & 1 & 1 \\ 1 & 1 & 1 \end{vmatrix}$=0, $\begin{vmatrix} 0 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 0 & 1 \end{vmatrix}$=0, $\begin{vmatrix} 1 & 0 \\ 2 & 1 \end{vmatrix}$=1≠0

Therefore Rank(A)=2.

Find the rank of the matrix $\mathbf{A} = \begin{bmatrix} \mathbf{2} & \mathbf{3} & \mathbf{-1} & \mathbf{-1} \\ \mathbf{1} & \mathbf{-1} & \mathbf{-2} & \mathbf{-4} \\ \mathbf{3} & \mathbf{1} & \mathbf{3} & \mathbf{-2} \\ \mathbf{6} & \mathbf{3} & \mathbf{0} & \mathbf{-7} \end{bmatrix}$

Solution:

$$\mathrm{A} = \begin{bmatrix} 2 & 3 & -1 & -1 \\ 1 & -1 & -2 & -4 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 3 & -1 & -1 \\ 1 & -1 & -2 & -4 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & -2 & -4 \\ 2 & 3 & -1 & -1 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{bmatrix} R_1 <\!\!-\!\!> R_2$$

$$\mathrm{A} \sim \begin{bmatrix} 1 & -1 & -2 & -4 \\ 0 & 5 & 3 & 7 \\ 0 & 4 & 9 & 10 \\ 0 & 9 & 12 & 17 \end{bmatrix} R_2 \!-\!\!> R_2 - 2R_1,\ R_3 \!-\!\!> R_3 - 3R_1,\ R_4 \!-\!\!> R_4 - 6R_1$$

$$A \sim \begin{bmatrix} 1 & -1 & -2 & -4 \\ 0 & 5 & 3 & 7 \\ 0 & 0 & \frac{33}{5} & \frac{22}{5} \\ 0 & 0 & \frac{33}{5} & \frac{22}{5} \end{bmatrix} R_3 \rightarrow R_3 - \frac{4}{5} R_2 \;, R_4 \rightarrow R_4 - \frac{9}{5} R_2$$

$$A \sim \begin{bmatrix} 1 & -1 & -2 & -4 \\ 0 & 5 & 3 & 7 \\ 0 & 0 & \frac{33}{5} & \frac{22}{5} \\ 0 & 0 & 0 & 0 \end{bmatrix} R_4 \rightarrow R_4 - R_3$$

Which is in Echelon form. The number of non-zeros rows is 3.

Therefore, the **rank(A)=3**

Find the rank of the matrix by converting it into Normal form $\boldsymbol{A} = \begin{bmatrix} \mathbf{-1} & \mathbf{2} & \mathbf{3} & \mathbf{-2} \\ \mathbf{2} & \mathbf{-5} & \mathbf{1} & \mathbf{2} \\ \mathbf{3} & \mathbf{-8} & \mathbf{5} & \mathbf{2} \\ \mathbf{5} & \mathbf{-12} & \mathbf{-1} & \mathbf{6} \end{bmatrix}$

- Solution:

$$A = \begin{bmatrix} -1 & 2 & 3 & -2 \\ 2 & -5 & 1 & 2 \\ 3 & -8 & 5 & 2 \\ 5 & -12 & -1 & 6 \end{bmatrix}$$

$$A \sim \begin{bmatrix} 1 & -2 & -3 & 2 \\ 2 & -5 & 1 & 2 \\ 3 & -8 & 5 & 2 \\ 5 & -12 & -1 & 6 \end{bmatrix} R_1 \text{—>} (-1)\, R_1$$

$$A \sim \begin{bmatrix} 1 & -2 & -3 & 2 \\ 0 & -1 & 7 & -2 \\ 0 & -2 & 14 & -4 \\ 0 & -2 & -14 & -4 \end{bmatrix} R_2 \text{—>} R_2 - 2R_1, R_3 \text{—>} R_3 - 3R_1, R_4 \text{—>} R_4 - 5R_1$$

$$A \sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 7 & -2 \\ 0 & -2 & 14 & -4 \\ 0 & -2 & 14 & -4 \end{bmatrix} C_2 \text{—>} C_2 + 2C_1, C_3 \text{—>} C_3 + 3C_1, C_4 \text{—>} C_4 - 2C_1$$

$$A \sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 7 & -2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} R_3 \text{—>} R_3 - 2R_2, R_4 \text{—>} R_4 - 2R_2$$

$$A \sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} \boldsymbol{I_2} & \boldsymbol{O} \\ \boldsymbol{O} & \boldsymbol{O} \end{bmatrix} C_3 \text{—>} C_3 + 7C_2, C_4 \text{—>} C_4 - 2C_2, C_2 \text{—>} (-1)C_2$$

$Therefore,$ $\boldsymbol{Rank(A) = 2}$

Observations

1. $Rank\ of\ A\ and\ A'\ is\ same.$
2. $Rank\ of\ null\ matrix\ is\ zero.$
3. For a rectangular matrix A of order m x n, $\text{Rank(A)} \leq \min\{m, n\}$ $i.e, rank\ can\ not\ exceed\ the\ smaller\ of\ m\ and\ n.$
4. For square matrix of order n if $\text{rank(A)} = \text{n then } |\text{A}| \neq 0 \text{i. e, A is non} - \text{singular.}$
5. For any square matrix of order n, $\text{if rank} < \text{n, then } |\text{A}| = 0 \text{ i. e, A is singular.}$

Practice Problems

1. Find the inverse of the matrix using Gauss-Jordan method

$$i) = \begin{bmatrix} 2 & -6 & -2 & -3 \\ 5 & -13 & -4 & -7 \\ -1 & 4 & 1 & 2 \\ 0 & 1 & 0 & 1 \end{bmatrix} \quad ii)A = \begin{bmatrix} 1 & 3 & 3 \\ 1 & 4 & 3 \\ 1 & 3 & 4 \end{bmatrix} \quad iii)A = \begin{bmatrix} 2 & 1 & -1 & 2 \\ 1 & 3 & 2 & -3 \\ -1 & 2 & 1 & -1 \\ 2 & -3 & -1 & 4 \end{bmatrix}$$

2. Find the rank of the matrix by converting it into Normal form

$$i)A = \begin{bmatrix} 1 & 2 & 3 & 2 \\ 2 & 3 & 5 & 1 \\ 1 & 3 & 4 & 5 \end{bmatrix} \quad ii)A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \\ 3 & 1 & 2 \end{bmatrix} \quad iii)A = \begin{bmatrix} 3 & 0 & 2 & 2 \\ -6 & 42 & 24 & 54 \\ 21 & -21 & 0 & 15 \end{bmatrix}$$

3.Determine the non-singular matrices P and Q such that PAQ is in the normal form for A . Hence find the rank of A

$$i)A = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 3 \\ 0 & -1 & -1 \end{bmatrix} \quad ii)A = \begin{bmatrix} 1 & 2 & 3 & -2 \\ 2 & -2 & 1 & 3 \\ 3 & 0 & 4 & 1 \end{bmatrix} \quad iii)A = \begin{bmatrix} 1 & -1 & 2 & -1 \\ 4 & 2 & -1 & 2 \\ 2 & 2 & -2 & 0 \end{bmatrix}$$

4.Find the rank of the following matrices

i) $A = \begin{bmatrix} 1 & 1 & -1 & 1 \\ 1 & -1 & 2 & -1 \\ 3 & 1 & 0 & 1 \end{bmatrix}$ $ii) A = \begin{bmatrix} 1 & 2 & 3 & 2 \\ 2 & 3 & 5 & 1 \\ 1 & 3 & 4 & 5 \end{bmatrix}$ $iii) A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 4 & 2 \\ 2 & 6 & 5 \end{bmatrix}$